

Q.1

Suppose $U_0 \in T$ is a limit point of T , G is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then, show that $f \circ G$ is continuous at U_0 ? Mark: 10

Proof:-

Suppose $\varepsilon > 0$. Since f is continuous at $X_0 = G(U_0)$, there is an $\varepsilon_1 > 0$ such that $|f(X) - f(G(U_0))| < \varepsilon \implies$ ① if

$$|X - G(U_0)| < \varepsilon_1 \text{ and } X \in D_f \implies$$

since G is continuous at U_0 , there is a $\delta > 0$ such that

$$|G(U) - G(U_0)| < \varepsilon_1 \quad \text{N.M.}$$

if $|U - U_0| < \delta$ and $U \in D_G$.

by taking $X = G(U)$ in ① and ② we get,

$$|h(U) - h(U_0)| = |f(G(U)) - f(G(U_0))| < \varepsilon$$

if

$$|U - U_0| < \delta \text{ and } U \in T$$

Note:-

“U” is a sign in Math type.

Q.2

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0)$?

Soln:-

Given $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$

$$p_x(x,y) = 2x - 2y - 4x^3$$

$$p_y(x,y) = -2x + 2y - 4y^3$$

At point $(0,0)$ M.M

$$p_x(0,0) = 0 \quad \text{and} \quad p_y(0,0) = 0$$

So, $(0,0)$ is a critical point of p .

Now,

$$p_{xx}(x,y) = 2 - 12x^2$$

$$p_{yy}(x,y) = 2 - 12y^2$$

$$p_{xy}(x,y) = -2$$

$$(d^2_{(0,0)} p)(x,y) = 2x^2 - 4xy + 2y^2$$

$$\begin{aligned} (d^2_{(0,0)} p)(x,y) &= 2(x^2 - 2xy + y^2) \\ &= 2(x-y)^2 \end{aligned}$$

Hence it is the semidefinite.

Q.3

Show that $(0,0)$ is a critical point of the function $P(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$ and if has the semidefinite second differentials at $(0,0)$.

Soln:-

Given

$$P(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$$

$$P_x(x,y) = 2x - 2y + 4x^3$$

$$P_y(x,y) = -2x + 2y + 4y^3$$

At point $(0,0)$

$$P_x(0,0) = 0 \quad \text{and} \quad P_y(0,0) = 0$$

So $(0,0)$ is a critical point of P .

Now

$$P_{xx}(x,y) = 2 + 12x^2$$

$$P_{yy}(x,y) = 2 + 12y^2$$

$$P_{xy}(x,y) = -2$$

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$$(d^2_{(0,0)}P)(x,y) = 2x^2 - 4xy + 2y^2$$

$$(d^2_{(0,0)}P)(x,y) = 2(x^2 - 2xy + y^2)$$

$$(d^2_{(0,0)}P)(x,y) = 2(x-y)^2$$

Hence it is the semidefinite.

Q.4

Define a locally integrable function.

Also show that the function $f(x) = \log x$ is locally integrable on $[0, 1]$.

Soln:-

Locally integrable function.

If f is a function, it is locally integrable on an interval I if f is integrable on every finite closed subinterval of I .

Proof:-

$$\int_1^0 \log x \, dx = x \ln x - x \Big|_1^0 \quad \text{M.M.}$$

$$= 0 - 1$$

$$= -1$$

Because it is unbounded as $x \rightarrow 0^+$

since

$$\lim_{c \rightarrow 0^+} \int_c^1 \log x \, dx = \lim_{c \rightarrow 0^+} (x \log x - x) \Big|_c^1$$

$$= -1 - \lim_{c \rightarrow 0^+} (c \log c - c) = -1$$

Q-5

Suppose that $f(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational} \\ 3 & \text{if } x \text{ and } y \text{ are irrational} \end{cases}$

Soln:-

Let $P = \{R_1, R_2, \dots, R_K\}$

$$M_j = \sup \{f(x,y) \mid x \in R_j\}$$

$$m_j = \inf \{f(x,y) \mid x \in R_j\}$$

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Let j be arbitrary in $\{1, 2, \dots, K\}$. Since R_j contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational

$M_j = 3$. Hence, $\int_R f(x,y) d(x,y) = 3(b-a)(d-c)$

Since R_j contains a point $(\tilde{x}_j, \tilde{y}_j)$ with

$\tilde{x}_j$ and $\tilde{y}_j$ rational. $m_j = 0$. Hence,

$$\underline{\int_R} f(x,y) d(x,y) = 0$$

Q.6

If a nonempty subset S of $\mathbb{R}^n$ is both open and closed, then $S = \mathbb{R}^n$.

Soln:-

Let x_0 is in S . If $S \neq \mathbb{R}^n$, there is an $x_1 \notin S$.

Let $H = \{t \mid (1-t)x_0 + tx_1 \in S \text{ for } 0 \leq t < 1\}$

Since S is open it contains a neighborhood of x_0 . So, $H \neq \emptyset$.

Since x_1 is not in S , $t \leq 1$ for all t in H .

Let $\rho = \sup H$ and $\bar{x} = (1-\rho)x_0 + \rho x_1$

then $\bar{x}$ is a limit point of S and so

in S . because S is closed. since S is open.

it contains some ϵ -neighborhood of $\bar{x}$. So

$\rho + \frac{\epsilon}{2}$ is in H . It gives the definition of

ρ .

Hence, $S = \mathbb{R}^n$.

M.M

Q.7

Define Linear Function:-

Ans:-

A linear function is a form of function

$$L(x) = m_1x_1 + m_2x_2 + \dots + m_nx_n$$

where $m_1, m_2, \dots, m_n$ are constant.

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Q.8

If R is degenerate, then prove by using the definition of Riemann integral that $\int_R f(x) dA = 0$ if f is bounded on R .

Proof:-

Let

$$R = [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n] \text{ with}$$

$$a_1 = b_1$$

then every partition P of R is of the

form

$$P = [R_1, R_2, R_3, \dots, R_K] \text{ where } \cancel{R_1 = \emptyset}$$

$$1 \leq j \leq K.$$

Therefore,

Every Riemann sum of f over R equals

Zero, So

$$\int_R f(x) dA = 0$$

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Q.9

Determine whether the integral

$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1+x^2}} dx$$

converges or diverges.

Sol:-

Since

$$\frac{1 + \cos^2 x}{\sqrt{1+x^2}} > \frac{1}{\sqrt{1+x^2}} = \frac{1}{x\sqrt{\frac{1}{x^2} + 1}} > \frac{1}{x\sqrt{2}}$$

if $x > 1$.

Since M.M $\nearrow$

$$\int_1^{\infty} \frac{dx}{x} = \infty$$

So,

$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1+x^2}} = \infty$$

this is diverges.

Q.10

Calculate $\frac{\partial f(x)}{\partial \phi}$ where $f(x, y, z) = \log(1+x)$

and $\phi = (0, 1, 0)$.

Soln-

We know that

$$h(t) = f(x + t\phi) \quad \text{M.M.}$$

Given

$$f(x, y, z) = \log(1+x+y+z)$$

$$\begin{aligned} h(t) &= f(x + \phi_1 t, y + \phi_2 t, z + \phi_3 t) \\ &= \log(1+x+y+z + (\phi_1 + \phi_2 + \phi_3)t) \end{aligned}$$

$$h'(t) = \frac{\phi_1 + \phi_2 + \phi_3}{1+x+y+z + (\phi_1 + \phi_2 + \phi_3)t}$$

at $t=0$ we get,

$$\frac{\partial f(x)}{\partial \phi} = h'(0) = \frac{\phi_1 + \phi_2 + \phi_3}{1+x+y+z}$$

Now put $\phi_1=0, \phi_2=1, \phi_3=0$

$$\frac{\partial f(x)}{\partial \phi} = \frac{0+1+0}{1+x+1+z} = \frac{1}{1+x+y+z}$$

Q.11

Prove that the function $f(x,y) = 3x^2y^3 + xy$ has eight third-order partial derivatives.

Sol.:

Given $f(x,y) = 3x^2y^3 + xy$

1st-order partial derivatives

$$f_x(x,y) = 6xy^3 + y$$

$$f_y(x,y) = 9x^2y^2 + x$$

2nd-order partial derivatives

$$f_{xx}(x,y) = 6y^3$$

$$f_{yy}(x,y) = 18x^2y$$

$$f_{xy}(x,y) = 18xy^2 + 1$$

$$f_{yx}(x,y) = 18xy^2 + 1$$

There are eight third-order derivatives.

$$f_{xxx}(x,y) = 0$$

$$f_{yyy}(x,y) = 18x^2$$

$$f_{xyy}(x,y) = 18y^2$$

$$f_{yyx}(x,y) = 18y^2$$

$$f_{yxx}(x,y) = 18y^2$$

$$f_{yxy}(x,y) = 18y^2$$

$$f_{xyx}(x,y) = 18y^2$$

$$f_{xxy}(x,y) = 18y^2$$

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Q.12

Compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$

if $f(x,y) = \begin{cases} \frac{(x^2y + xy^2) \sin(x-y)}{x^2 + y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

sol:-

$$f_{xy}(0,0) = \lim_{y \rightarrow 0} \frac{f_x(0,y) - f_x(0,0)}{y}$$

$$= \lim_{y \rightarrow 0} \frac{-\sin y - 0}{y} = -1$$

$$f_{yx}(0,0) = \lim_{y \rightarrow 0} \frac{f_y(0,y) - f_y(0,0)}{y}$$

$$= \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0$$

M.M

Q.13

Maxima and Minima

Ans:-

If X_0 is a local extreme point of f if there is a $\delta > 0$ such that

$$f(x) - f(X_0)$$

does not change sign in $S_\delta(X_0) \cap D_f$

then Maxima is $f(x) \leq f(X_0)$

and minima is $f(x) \geq f(X_0)$ for all x in $S_\delta(X_0) \cap D_f$

Q.14

Show that the function $f(x, y) = x^2 + 2xy$ is differentiable at point (x_0, y_0) .

Sol:-

$$f(x, y) - f(x_0, y_0) = x^2 + 2xy - x_0^2 - 2x_0y_0$$

$$= x^2 - x_0^2 + 2(xy - x_0y_0)$$

$$\text{N.M.} \quad = (x - x_0)(x + x_0) + 2(xy - x_0y_0)$$

$$= (x - x_0)(x + x_0) + 2(xy - x_0y_0 + x_0y - x_0y_0)$$

$$= (x - x_0)(x + x_0) + 2(xy - x_0y_0) + 2(x_0y - x_0y_0)$$

$$= (x - x_0)(x + x_0) + 2y(x - x_0) + 2x_0(y - y_0)$$

$$= (x - x_0)(x + x_0 + 2y) + 2x_0(y - y_0)$$

$$= 2(x_0 + y_0)(x - x_0) + 2x_0(y - y_0) + (x - x_0)$$

$$(x - x_0 + 2y - 2y_0)$$

$$= m_1(x - x_0) + m_2(y - y_0) + (x - x_0)(x - x_0 + 2y - 2y_0)$$

here $m_1 = 2(x_0 + y_0) = f_x(x_0, y_0)$ and $m_2 = 2x_0 = f_y(x_0, y_0)$

~~hence~~ by Schwarz's inequality

$$\lim_{x \rightarrow x_0} \frac{f(x, y) - f(x_0, y_0) - m_1(x - x_0) - m_2(y - y_0)}{|x - x_0|} = 0$$

So, function is differentiable at (x_0, y_0) .

Q.15

Define Riemann sum and Riemann integral in $\mathbb{R}^n$.

Ans:—

Riemann sum in $\mathbb{R}^n$

If f is a real-valued function defined on a rectangle R in $\mathbb{R}^n$ and $P = \{R_1, R_2, \dots, R_k\}$ is a partition of R , let X_j is an arbitrary point in R_j such that $1 \leq j \leq k$

M.M

$$\sigma = \sum_{j=1}^k f(X_j) V(R_j)$$

is a Riemann sum of f over P .

Riemann integral in $\mathbb{R}^n$

If σ is any Riemann sum of f over a partition P of R such that $\|P\| < \delta$. here, we can say that L is the Riemann integral of f over R and written as

$$\int_R f(x) dx = L$$